

Final Project – Checkpoint B

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Introduction

This project studies whether an actively managed equity fund focused on the nuclear power sector can deliver superior risk-adjusted returns relative to passive nuclear exposure. With the forecasted growth of AI, data centers, and quantum computing, electricity demand is rising faster than at any point in recent decades, and companies powering these workloads will need to source energy from cleaner, more scalable alternatives to traditional fossil fuels. This fund is built on the thesis that nuclear power will become the dominant source of baseload electricity for large-scale computing infrastructure, driven by its ability to deliver continuous 24/7 generation, its high energy density, and its low-carbon profile relative to coal and natural gas. Evidence of this shift can be found in recent agreements between major technology firms and nuclear operators.

The fund's holdings will be segmented into 3 groups within the nuclear space: uranium mining, next-generation reactors, and utilities with nuclear generation capacity. The fund applies Markowitz mean variance optimization for bucket weighting, a momentum-based trading overlay for position sizing, and Monte Carlo simulation for forward return analysis.

The fund's holdings are composed of a mixed group of equities with varying levels of historical data to analyze. We will have to both utilize historical analysis for those equities where there is enough data as well as properly anticipate the future energy needs via python libraries like GDELT, and Finnhub.

Literature Review

Recent literature suggests that nuclear energy is entering a period of renewed global importance driven by energy security concerns, decarbonization goals, and rising electricity demand from AI infrastructure and industrial electrification. This trend is particularly relevant to the development of the nuclear-focused investment fund being proposed, as many of the underlying macroeconomic and geopolitical themes support long-term growth in uranium demand, reactor construction, and nuclear-related equities.

The International Energy Agency's report, *The Path to a New Era for Nuclear Energy*, argues that nuclear power is increasingly viewed as a necessary component of the global energy

transition. The report highlights how governments are reevaluating nuclear energy due to the need for reliable baseload power and reduced dependence on fossil fuels. The IEA projects significant long-term growth in nuclear investment and capacity expansion through 2050, especially in advanced economies seeking both carbon reduction and energy independence. The articles projects global nuclear capacity to rise by over 50% by 2050 (IEA 2025). For the proposed nuclear fund, this supports the idea that nuclear energy is transitioning from a niche sector into a strategically important industry with growing institutional and governmental backing. This macro trend could create sustained capital inflows into uranium miners, utilities, and reactor technology companies.

Similarly, the *World Nuclear Outlook Report* published by the World Nuclear Association reinforces the bullish long-term demand outlook for the sector. The report projects substantial increases in global nuclear generating capacity under both reference and optimistic scenarios. It also emphasizes that many countries are extending the life of existing reactors while simultaneously investing in new builds and small modular reactor (SMR) technologies. This is particularly relevant from an investment perspective because reactor expansion directly increases long-term uranium demand and strengthens the outlook for firms involved in the nuclear fuel cycle. For the fund, the report supports an investment thesis centered on structural growth rather than short-term commodity speculation. Companies involved in uranium mining, enrichment, and nuclear infrastructure may benefit from multi-decade demand tailwinds if these forecasts materialize.

For the proposed nuclear fund, the literature supports a long-term investment strategy focused on uranium producers, reactor developers, nuclear utilities, and firms positioned to benefit from increased global nuclear adoption. While risks such as regulatory delays and cost overruns remain important, the overall academic and industry outlook increasingly points toward sustained growth in the nuclear energy sector over the coming decades.

Methods

The fund consists of eleven equities across three buckets:

- *Uranium miners and fuel cycle:* Cameco (CCJ), Uranium Energy (UEC), NexGen Energy (NXE), Denison Mines (DNN)
- *Reactor developers:* Oklo (OKLO), NuScale Power (SMR), Nano Nuclear Energy (NNE), Centrus Energy (LEU)
- *Utilities with nuclear generation:* Vistra (VST), Constellation Energy (CEG), Talen Energy (TLN)

The primary benchmark is the Global X Uranium ETF (URA), with secondary benchmarks of the Energy Select Sector SPDR (XLE) and the S&P 500 (SPY) to evaluate the fund against both sector and broad-market exposures.

Daily adjusted closing prices will be sourced from Yahoo Finance through the yfinance Python library, with QuantConnect used as a secondary source for backtesting infrastructure. Maximum available history will be used for each name, with the recognition that several reactor developers have short trading histories (Oklo listed in May 2024, Nano Nuclear in 2023). Fundamental data, regulatory headlines, and ESG indicators will be sourced from Finnhub. Alpha Vantage will support the backtesting framework. For sentiment and event-based features, FinBERT will be used to score financial news, and GDELT will provide a broader feed for tracking government subsidies, regulatory approvals, and policy events related to nuclear power.

Total returns will be computed at daily frequency using adjusted closing prices. Log returns will be used for statistical modeling and volatility estimation, while simple returns will be used for performance reporting. Key variables include rolling 60-day and 252-day volatility, correlation to URA, and beta versus URA, XLE, and SPY.

Modeling Methods

- Markowitz mean-variance optimization will be used to set initial weights across the three buckets, using historical covariance estimated over the longest common data window.
- Monte Carlo simulation will generate forward return distributions for the portfolio under base, bull, and bear assumptions about AI-driven power demand growth.
- Moving average crossover signals (50-day and 200-day) will serve as the primary momentum overlay for position sizing within each bucket.
- Algorithmic trading system design will specify rebalancing rules buy and sell triggers based on the moving average signal, and execution logic suitable for systematic implementation.

Results

Data availability has been confirmed for the longer-history names in the universe. Cameco, Constellation, and Vistra all have multi-year daily price histories suitable for full backtesting of the moving average signal. The uranium miners as a group have sufficient history to evaluate sleeve-level performance against URA. Data limitations *have been identified* for the reactor developer bucket. Oklo, Nano Nuclear, and Centrus have short trading histories that limit the reliability of historical backtesting.

Initial observations on returns and volatility show that the nuclear universe has exhibited substantially higher volatility than the broad market over the trailing two-year window, with uranium miners and reactor developers showing the highest realized volatility and utilities

the lowest. Preliminary correlation analysis indicates that the three buckets are positively but imperfectly correlated, supporting the diversification premise of the three-sleeve structure, though the correlations tighten meaningfully during sector-wide drawdowns.

Exploratory analysis of event signals using GDELT indicates that major PPA announcements and regulatory approvals over the past two years are associated with measurable short-term price moves in utility names with direct exposure, supporting the inclusion of event-based features in the model.

Work Cited -

International Energy Agency. *The Path to a New Era for Nuclear Energy*. Paris: International Energy Agency, 2025. [IEA Report](#)

World Nuclear Association. *World Nuclear Outlook Report*. London: World Nuclear Association, 2025. [World Nuclear Outlook Report](#)